Explaining the STAR Case in the Strips Proof for Version 2 of `shifts`

Goal

We want to explain the STAR case of the correctness proof for version 2 of `shifts`. The statement being proved is:

$$P(ms, r) : \text{set}(\text{shifts}\ \ ms\ r) = \text{Strips}(L(r))(\text{set}\ \ ms).$$

In the STAR case, this becomes:

$$\text{set}(\text{shifts}\ \ ms\ r^*) = \text{Strips}(L(r^*))(\text{set}\ \ ms).$$

The point of this note is to explain carefully:

- what the symbols mean,
- why the two STAR lemmas are true,
- what the key semantic identity means,
- and how the STAR proof uses those lemmas.

Basic notation

Sets and symbols

If we write

$$X \subseteq Y,$$

this means:

> every element of X is also an element of Y.

So X is contained in Y.

If we write

$$X = Y,$$

this means:

> X and Y have exactly the same elements.

A common way to prove $X = Y$ is to prove both

$$X \subseteq Y \quad \text{and} \quad Y \subseteq X.$$

If we write

$$X - Y,$$

this means the *set difference*: all elements that are in X but not in Y.

If we write

$$\emptyset,$$

this means the empty set, the set with no elements.

Why $X - Y = \emptyset$ means the same as $X \subseteq Y$

This is standard set theory.

The statement

$$X - Y = \emptyset$$

means:

> there is no element of X that lies outside Y.

But that is exactly the same as saying:

> every element of X lies inside Y.

And that is exactly the meaning of

$$X \subseteq Y.$$

So:

$$X - Y = \emptyset \quad \Longleftrightarrow \quad X \subseteq Y.$$

In our proof, this is used in the form

$$\mathrm{Strips}(A, B) - B = \emptyset \quad \Longleftrightarrow \quad \mathrm{Strips}(A, B) \subseteq B.$$

What $\mathrm{Strips}(A, B)$ means

The new Strips is defined from

$$\mathrm{Strip}(A, s) = \{\, s_2 \mid \exists s_1 \in A.\ s = s_1 @ s_2 \,\},$$

and then

$$\mathrm{Strips}(A, B) = \bigcup \big(\mathrm{Strip}(A)\, '\, B\big).$$

The important point is that the removed prefix s_1 is allowed to be empty. So if $[] \in A$, then a string in B may remain unchanged.

Intuitively:

> $\mathrm{Strips}(A, B)$ means: all strings you can get by removing one prefix from A from strings in B.

If $A = L(r)$, then:

- $\mathrm{Strips}(L(r), B)$ means one pass through r,
- $\mathrm{Strips}(L(r)^*, B)$ means zero or more passes through r.

Operational STAR case

Version 2 of `shifts` defines STAR as:

$$\text{shifts } ms\ (r^*) = \text{let } ms^{\#} = (\text{shifts } ms\, r) - ms \text{ in if } ms^{\#} = [] \text{ then } ms \text{ else } ms @ \text{shifts } ms^{\#}\ (r^*).$$

This means:

1. shift once through r,
2. remove the marks that did not really change,

3. call the genuinely new shorter marks $ms^{\#}$,
4. if no such marks exist, stop and return ms,
5. otherwise keep ms and recurse on $ms^{\#}$.

So the proof naturally splits into two cases:

- $(\text{shifts } ms\ r) - ms = []$,
- $(\text{shifts } ms\ r) - ms \neq []$.

Lemma 1

Statement

If

$$\text{Strips}(A, B) - B = \emptyset,$$

then

$$\text{Strips}(A^*, B) = B.$$

Since

$$\text{Strips}(A, B) - B = \emptyset \quad \Longleftrightarrow \quad \text{Strips}(A, B) \subseteq B,$$

we can also read this lemma as:

> if one A-step never leaves B, then any finite number of A-steps never leaves B, and because star also allows zero steps, the result is exactly B.

Why this lemma is true

We prove first:

$$\text{if Strips}(A, B) \subseteq B, \text{ then for all } n, \text{ Strips}(A^n, B) \subseteq B.$$

Base case $n = 0$. Since $A^0 = \{[]\}$, stripping by A^0 changes nothing, so

$$\text{Strips}(A^0, B) = B.$$

Hence

$$\text{Strips}(A^0, B) \subseteq B.$$

Induction step. Assume

$$\text{Strips}(A^n, B) \subseteq B.$$

We want

$$\text{Strips}(A^{n+1}, B) \subseteq B.$$

Using the concatenation property of Strips,

$$\text{Strips}(A^{n+1}, B) = \text{Strips}(A, \text{Strips}(A^n, B)).$$

By the induction hypothesis,

$$\text{Strips}(A^n, B) \subseteq B.$$

By monotonicity of Strips in its second argument,

$$X \subseteq Y \implies \mathrm{Strips}(A, X) \subseteq \mathrm{Strips}(A, Y),$$

we get

$$\mathrm{Strips}(A, \mathrm{Strips}(A^n, B)) \subseteq \mathrm{Strips}(A, B).$$

Now use the assumption

$$\mathrm{Strips}(A, B) \subseteq B.$$

So

$$\mathrm{Strips}(A^{n+1}, B) \subseteq B.$$

Thus the claim holds for all n.

From powers to star. Now use

$$A^* = \bigcup_{n \geq 0} A^n.$$

Therefore

$$\mathrm{Strips}(A^*, B) = \mathrm{Strips}\Big(\bigcup_{n \geq 0} A^n,\ B\Big) = \bigcup_{n \geq 0} \mathrm{Strips}(A^n, B).$$

Each set $\mathrm{Strips}(A^n, B)$ is contained in B, so their union is also contained in B. Hence

$$\mathrm{Strips}(A^*, B) \subseteq B.$$

On the other hand,

$$B \subseteq \mathrm{Strips}(A^*, B),$$

because $[] \in A^*$, so stripping the empty string is always allowed and leaves every element of B unchanged.

So we have both inclusions:

$$\mathrm{Strips}(A^*, B) \subseteq B \quad \text{and} \quad B \subseteq \mathrm{Strips}(A^*, B).$$

Hence

$$\mathrm{Strips}(A^*, B) = B.$$

Meaning of Lemma 1

Lemma 1 says:

> if one pass through A never produces anything outside B, then repeated passes through A also never produce anything outside B; since star also includes zero passes, star gives exactly B.

This is exactly what is needed in the easy STAR case.

Lemma 2: the key semantic identity

Statement

The second lemma is the identity

$$\mathrm{Strips}(A^*, B) = B \cup \mathrm{Strips}(A^*)\big(\mathrm{Strips}(A, B) - B\big). \tag{†}$$

What "semantic identity" means

Here "semantic identity" means:

> a mathematically true equality about the semantic operator Strips.

It is called semantic because Strips describes what strings remain after matching prefixes from a language. So it talks about the meaning of the regex, not about the syntax of the program.

This identity is not just guessed. It is a set-theoretic property that the proof establishes using lemmas about powers A^n, monotonicity, and the decomposition of A^*.

What the identity means intuitively

The left-hand side

$$\mathrm{Strips}(A^*, B)$$

means all results obtainable from B by doing zero or more A-steps.

The right-hand side says that every such result is of one of two kinds:

1. it is already in B, corresponding to taking zero A-steps;
2. or it comes from making one genuinely new A-step and then continuing with zero or more further A-steps.

Why do we write

$$\mathrm{Strips}(A, B) - B$$

instead of just $\mathrm{Strips}(A, B)$?

Because in the new version, empty stripping is allowed. So after one A-step, some results may still already be in B. The term

$$\mathrm{Strips}(A, B) - B$$

keeps only the genuinely new results, the ones that really moved forward.

This exactly matches the operational definition

$$ms^{\#} = (\text{shifts } ms\ r) - ms.$$

How the identity is proved

To prove

$$\mathrm{Strips}(A^*, B) = B \cup \mathrm{Strips}(A^*)\big(\mathrm{Strips}(A, B) - B\big),$$

we prove both inclusions.

Easy direction

First show

$$B \cup \mathrm{Strips}(A^*)\big(\mathrm{Strips}(A, B) - B\big) \subseteq \mathrm{Strips}(A^*, B).$$

This has two parts.

Part 1.

$$B \subseteq \mathrm{Strips}(A^*, B),$$

because $[] \in A^*$.

Part 2.

$$\text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big) \subseteq \text{Strips}(A^*,B).$$

Why? Because

$$\text{Strips}(A,B)-B \subseteq \text{Strips}(A,B),$$

and anything obtained by one A-step from B, followed by zero or more extra A-steps, is certainly obtainable by zero or more total A-steps from B.

So the easy inclusion holds.

Hard direction

Now show

$$\text{Strips}(A^*,B) \subseteq B \cup \text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big).$$

The file proves this first for powers:

$$\text{Strips}(A^n,B) \subseteq B \cup \text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big) \quad \text{for all } n.$$

Base case $n=0$. Since

$$\text{Strips}(A^0,B) = B,$$

the result is immediate.

Induction step. Assume

$$\text{Strips}(A^n,B) \subseteq B \cup \text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big).$$

Then

$$\text{Strips}(A^{n+1},B) = \text{Strips}(A,\text{Strips}(A^n,B)).$$

By monotonicity,

$$\text{Strips}(A,\text{Strips}(A^n,B)) \subseteq \text{Strips}\Big(A,\ B \cup \text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big)\Big).$$

Distribute over union:

$$= \text{Strips}(A,B) \cup \text{Strips}\Big(A,\text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big)\Big).$$

Now use the auxiliary fact

$$\text{Strips}(A,\text{Strips}(A^*,C)) \subseteq \text{Strips}(A^*,C). \tag{$*$}$$

So

$$\text{Strips}\Big(A,\text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big)\Big) \subseteq \text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big).$$

Hence

$$\text{Strips}(A^{n+1},B) \subseteq \text{Strips}(A,B) \cup \text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big).$$

Finally, every element of $\text{Strips}(A,B)$ is either already in B, or lies in $\text{Strips}(A,B)-B$. So

$$\text{Strips}(A,B) \subseteq B \cup (\text{Strips}(A,B)-B).$$

And since A^* contains the empty word,

$$\text{Strips}(A,B)-B \subseteq \text{Strips}(A^*)\big(\text{Strips}(A,B)-B\big).$$

So
$$\mathrm{Strips}(A^{n+1}, B) \subseteq B \cup \mathrm{Strips}(A^*)\big(\mathrm{Strips}(A, B) - B\big).$$

This finishes the induction on n. Since
$$A^* = \bigcup_{n \geq 0} A^n,$$
the desired inclusion for A^* follows.

Thus the identity (†) is proved.

Meaning of the auxiliary fact $(*)$

The fact
$$\mathrm{Strips}(A, \mathrm{Strips}(A^*, C)) \subseteq \mathrm{Strips}(A^*, C)$$
means:

> if you first do zero or more A-steps, and then do one more A-step, you are still within the results of doing zero or more A-steps.

This is because
$$A^* = \{[]\} \cup A^*@A,$$
so one more A after an A^*-computation is still part of an A^*-computation.

Using the lemmas in the STAR proof

Now let
$$A = L(r), \qquad B = \mathrm{set}\ \ ms.$$

We want to prove
$$\mathrm{set}(\mathrm{shifts}\ \ ms\ r^*) = \mathrm{Strips}(L(r^*))(\mathrm{set}\ \ ms).$$

Case 1: $(\mathrm{shifts}\ \ ms\ r) - ms = []$

Operationally, version 2 of STAR returns
$$\mathrm{shifts}\ \ ms\ r^* = ms.$$

So the left-hand side is
$$\mathrm{set}(\mathrm{shifts}\ \ ms\ r^*) = \mathrm{set}\ \ ms = B.$$

By the induction hypothesis for r,
$$\mathrm{set}(\mathrm{shifts}\ \ ms\ r) = \mathrm{Strips}(L(r), B) = \mathrm{Strips}(A, B).$$

The assumption gives
$$\mathrm{Strips}(A, B) - B = \emptyset.$$

By Lemma 1,
$$\mathrm{Strips}(A^*, B) = B.$$

Since $A = L(r)$, this gives
$$\mathrm{Strips}(L(r^*), \mathrm{set}\ \ ms) = \mathrm{set}\ \ ms.$$

So the easy STAR case is proved.

Case 2: $(\text{shifts } ms\ r) - ms \neq []$

Operationally,

$$\text{set}(\text{shifts } ms\ r^*) = B \cup \text{set}(\text{shifts}((\text{shifts } ms\ r) - ms,\ r^*)).$$

By the induction hypothesis for the recursive STAR call,

$$\text{set}(\text{shifts}((\text{shifts } ms\ r) - ms,\ r^*)) = \text{Strips}(A^*)\big(\text{set}((\text{shifts } ms\ r) - ms)\big).$$

By the induction hypothesis for r,

$$\text{set}(\text{shifts } ms\ r) = \text{Strips}(A, B).$$

So

$$\text{set}(\text{shifts } ms\ r^*) = B \cup \text{Strips}(A^*)\big(\text{Strips}(A, B) - B\big).$$

Now apply Lemma 2:

$$\text{Strips}(A^*, B) = B \cup \text{Strips}(A^*)\big(\text{Strips}(A, B) - B\big).$$

Therefore

$$\text{set}(\text{shifts } ms\ r^*) = \text{Strips}(A^*, B) = \text{Strips}(L(r^*))(\text{set } ms).$$

So the second STAR case is proved.

Final intuition

The two lemmas say exactly this:

- **Lemma 1:** if one A-step never leaves B, then star also never leaves B, and because star allows zero steps, star gives exactly B.
- **Lemma 2:** every star-result is either already in B because zero steps were taken, or it comes from making one genuinely new A-step and then continuing with star.

That is why the semantic lemmas mirror the operational definition of STAR so closely.